



Research Article

Achieving anti-corrosion and anti-biofouling dual-function self-healing coating by natural carrier attapulgite loading with 2-Undecylimidazoline

Yinghao Wu^a, Jinming Wei^{a,*}, Xiaolu Shi^b, Wenjie Zhao^{a,*}^a Key Laboratory of Marine Materials and Related Technologies, Zhejiang Key Laboratory of Marine Materials and Protective Technologies, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo 315201, China^b Ningbo New Material Testing and Evaluation Center Co., Ltd, Ningbo 315048, China

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ABSTRACT

In the marine environment, the protective coatings face serious corrosion and biofouling problem, limitations, and challenges that made self-healing coatings unable to perform both anti-corrosion and anti-biofouling dual-function at the same time. Here we constructed the corrosion resistance and anti-biofouling self-healing coating by integrating nano-containers into the coating matrix, the 2-Undecylimidazoline (ULM) acted as a corrosion inhibitor and anti-biofouling dual-functional agent which was loaded on the natural container attapulgite (ATP). To obtain high healing efficiency, a fluidity-driven self-healing silicone oil coating was thickened by fibrous ATP to enhance its stability, which played a key role in the self-healing and long-term corrosion resistance. The self-healing time of ULM@ATP reinforced oil coating was 4 s at least in the air and up to 30 s in the 1 M HCl solution. Meanwhile, the ULM@ATP can significantly enhance the corrosion resistance of the self-healing coating, with the best effect achieved when the content of ULM@ATP was 5 wt.%. The impedance modulus of ULM@ATP-2 still reached $1.62 \times 10^8 \Omega \text{ cm}^2$ after 480 immersion in 3.5 wt.% NaCl solution, which is 3 orders of magnitude high than pure Oil coating ($2.17 \times 10^5 \Omega \text{ cm}^2$). The restructure of the ATP network and the release of ULM could largely inhibit the corrosion of metal. The continuous open circuit potential of the composite coating performed the potentially infinite self-healing capacity. The self-healing performances of the composite coating in strong acid and base solutions exhibited high environmental suitability. This anti-corrosion and anti-biofouling (the surface coverage of adhered chlorella decreased 96.88%) dual-function strategy of self-healing coatings could be also realized by many other porous nano-materials or function modifications. The self-healing performances of the composite coating in strong acid and base solutions exhibited high environmental suitability. This anti-corrosion and anti-biofouling dual-function strategy of self-healing coatings could be also realized by many other porous nano-materials or function modifications. The rapid response self-healing coating possessed anti-biofouling and active self-healing functions and showed wide application under more service environments.

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1. Introduction

Self-healing has attracted substantial interest in corrosion protective coatings because of its ability for healing damages, especially in harsh corrosive environments where the damage cannot be repaired manually [1]. The concept of self-healing coatings was first proposed by Jud et al. [2] to reduce the frequency of repairs and the associated costs, however, it was remaining an active re-

search topic over the past 40 years. Various technologies have been designed with new self-healing strategies and explored new self-healing mechanisms for enhancing self-healing functionality [3]. The evolution of self-healing materials (SHMs) is considered into four generations according to the healing mechanism [4] and features of the self-healing strategy [5] in the previous overview.

The first generation of self-healing materials was based on extrinsic mechanisms and employed encapsulated external healing agents [4]. Various micro/nanoencapsulation enhancements have been adopted to improve the self-healing effect of SHMs. The healing agents are stored in micro/nanocarriers [6] or microvascular networks [7], The healing agents are assumed to release and

* Corresponding authors.

E-mail addresses: weijinming@nimte.ac.cn (J. Wei), zhaowj@nimte.ac.cn (W. Zhao).

harden once the local fracture, crack, and scratch occurred. The second generation of self-healing material is based on the unique molecular architecture of the coatings, and thus, does not require any external component to be embedded [1]. The self-healing was based on reversible chemical bonding [8] such as hydrogen bonding [9], π - π stacking interaction [10], ionic interaction [11], and metal-ligand interaction [12]. The extrinsic healing agent is not needed, however, external stimulation such as heat, electricity [13], light [14], disulfide bond [15], and moisture [16] is required. The SHMs needing non-external stimuli with hydrogen bond, dynamic urea bond, or another dynamic bond are considered as the third generation. For instance, multiple hydrogen bonds showed room-temperature self-healing ability with no demand for external stimulus [17]. Above SHMs are limited greatly by time dependence, environmental stimuli, property recovery, reprocessing, and healing efficiency. Among all of these problems, minimizing the healing time is the most important and difficult which can enhance the healing efficiency and weaken the time dependence. One effective way to reduce the self-healing time is by conferring mobility to the material [4]. Fluid materials in nature are known to recover from damage rapidly without any external triggering, however, there are difficult to form a stable or reliable coating on metal substrates. Adding fillers into the fluid matrix can effectively improve the stability of the coating, which can deliver optimal mechanical properties [18] and enhance the viscosity [5] of the composite. For instance, the lightweight reduced graphene oxide microcapsule was added to the silicone oil to enhance its viscosity and performed a rapid self-healing property [19]. A stable and superfast self-healing coating was obtained by incorporating carbon nanotube into the fluid matrix of epoxy resin and silicone oil, excellent durability, and infinite self-healing capacity with short response time presented a potential application in corrosion barrier coating [5].

In the marine environment, the protective coatings face both corrosion and biofouling, which will result in protection degradation, huge economic loss, ecological environmental damage, or even catastrophic safety accidents. Therefore, the development of anti-corrosion and anti-biofouling coatings for marine engineering and offshore platform is of great significance to ensuring service safety and reliability. Until now, the studies of self-healing coatings mostly focused on only corrosion resistance or anti-biofouling property. Few self-healing coating research achieved both corrosion resistance and anti-biofouling in the same protective coating which can benefit wide application potential in the harsh marine environment. Thus, it is highly desirable to design and fabricate dual-function protective self-healing coatings with high environmental compatibility, long-term durability, and stability for anti-corrosion and anti-biofouling applications.

Herein, we present a dual-function self-healing coating with active corrosion protection and anti-biofouling performances, using 2-Undecylimidazoline (ULM) loaded on attapulgite (ATP) as a corrosion inhibitor and anti-biofouling agent. ATP is an abundant natural hydrated magnesium aluminum silicate with a high aspect ratio, one-dimensional nanorod crystal morphology, regularly arranged tunnels/channels, and rich surface active sites [20]. The abundant silanol groups (Si-OH) and the permanent charge generated by the isomorphous substitutions on the surface can act as active sites for interaction with other substances [21]. The rod-like morphology of ATP could be considered as the desired carrier [22], which is suitable for loading active nanoparticles, or other functional components of the nanocomposite. More importantly, as natural nanomaterials, the properties like high specific surface area, surface activity, and high porosity endow the ability and special behaviors of favorable carriers to the ATP which could be considered as the construct of innovative anti-microbial materials [23]. The antibacterial effect of ATP could be greatly improved after being loaded with nanoparticles or modified with organic com-

pounds [24]. The ULM loaded on treated ATP was used to inhibit corrosion [25] on the metal surface and anti-biofouling [26] when the coating was applied in the marine environment.

Besides self-healing ability, active corrosion protection ability is also a favorable property for the development of smart protective coatings used in harsh service environments where self-healing or replacement may be difficult and/or costly. Active corrosion protection coatings can repair corrosion damages and/or inhibit corrosion deterioration with minimal or no external intervention [27]. The active protection process of these smart coatings could be achieved by the release of healing agents or corrosion inhibitors [28] using various encapsulation technologies which could be conveniently released with triggering by external conditions. Imidazoline derivatives have been successfully used as corrosion inhibitors to prevent CO₂ and H₂S corrosion in oil-gas fields [25]. Imidazoline can form multi-center chemical absorption sites on the steel surface to provide satisfactory corrosion inhibition abilities [29]. Moreover, the imidazoline derivatives can act as a mixed-type inhibitor because they can adsorb to the steel surface and form a stable coating with strong adhesion [30].

In this work, the conceptual design of the self-healing coating is creative and different from conventional strategies. (1) The conventional self-healing coatings (e.g., needing external stimulus or with unitary function) only work single time at the same position and may decrease the protection of metal matrix due to their low healing efficiency and their incomplete self-healing properties. (2) The conventional coatings are generally taking a long time to heal the damaged areas, whereas the natural fluid oil matrix endows ULM@ATP/Oil with a high self-healing efficiency, facilitating fast damage healing speed. (3) The ULM functional molecule loaded on ATP could enhance the stability of the self-healing coating and realize the anti-corrosion and anti-biofouling properties at the same time. Consequently, the fluidity-driven self-healing coating possesses anti-biofouling and active self-healing functions, exhibiting high suitability for wide application and long-term service in various harsh environments.

2. Materials and methods

2.1. Materials

Attapulgite was purchased from Xuyi, Jiangsu (purity ~60%) with the chemical formula of (Al₂Mg₂)Si₈O₂₀(OH)₂·4H₂O. ULM was purchased from Beijing Bailing Wei Technology Co., Ltd. with analytical pure (AR). The ammonium fluoride (AR), silicone oil (AR), Sudan III (AR), ethanol (AR), and NH₃·H₂O (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd. Q235 low carbon steel (1 cm × 1 cm × 1 cm) was chosen as the metal substrate for work electrode in the electrochemical test. All other reagents were purchased from Aladdin industrial corporation with analytical pure and used directly. All solutions were prepared using degassed distilled water.

2.2. The preparation of ATP-based nanocontainer

The preparation process of the ATP-based nanocontainer is shown in Fig. 1. Firstly, 5 g of ATP and 1.5 g of ammonium fluoride were uniformly dispersed into 250 mL of deionized water and stirred for 4 h. The hole cutting carried out by ammonium fluoride was used to enhance the loading ability of ATP nanoparticles. After the reaction, the excess precipitation substances were washed with deionized water and then centrifuged at 9000 rpm min⁻¹ to eliminate the ammonium fluoride residue on the ATP surface. The procedure of washing and centrifuging was repeated three times. Finally, the ATP was freeze-dried under a vacuum for 24 h to obtain hole-cut ATP.

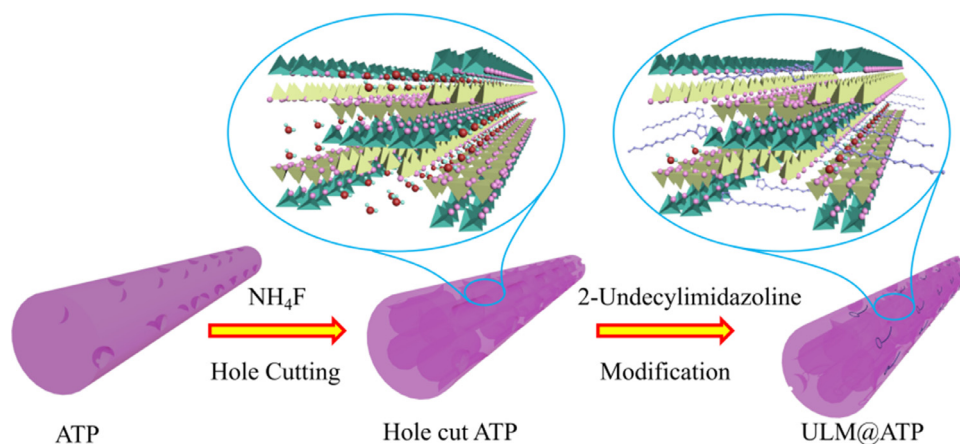


Fig. 1. Schematic representation of the synthesis for ULM@ATP.

2.3. The loading of ULM

Based on the porous structure of hole-cut ATP, ULM molecules were loaded into the ATP nano-container. Firstly, 5 g of hole-cut ATP, 1 g of undecyl imidazoline, and 250 ml of ethanol were mixed and vigorously stirred for 10 h. After the reaction, the excess precipitation substances were rinsed via deionized water and then centrifuged at $9000 \text{ rpm min}^{-1}$ to eliminate the redundant ULM on the ATP surface. The procedure of washing and centrifuging was repeated three times. Finally, the product was freeze-dried under a vacuum for 24 h to obtain 2-undecylimidazoline@attapulgit (ULM@ATP) nanomaterials (Fig. 1).

2.4. Preparation of ULM@ATP/Oil coating

The ULM@ATP was dispersed into the silicone oil via ultrasonic oscillation for 1 h. For convenience, the coating with 2.5 wt.%, 5.0 wt.%, and 7.5 wt.% ULM@ATP were labeled as ULM@ATP/Oil-1, ULM@ATP/Oil-2, and ULM@ATP/Oil-3, respectively. Similarly, the coating with 2.5 wt.%, 5.0 wt.%, and 7.5 wt.% ATP were labeled as ATP/Oil-1, ATP/Oil-2, ATP/Oil-3, respectively.

2.5. Structure characterization

To study the structure and composition of ATP, the hole-cut ATP, and ULM@ATP, Fourier transform infrared spectrometer (FTIR, Nicolet 6700, USA), Raman spectrometer (Renishaw inVia Reflex, Renishaw, England) and X-ray diffractometer (XRD, D8 ADVANCE, UK) were used. Raman spectrometer was equipped with a 532 nm laser and the spot size was $1 \mu\text{m}$. The Thermo Fisher electron spectrometer (K-Alpha) with an Al $K\alpha$ X-ray source was used (excitation energy = 160 eV). Field emission scanning electron microscopy (SEM, FEI Quanta 250 FEG, USA) and transmission electron microscopy (TEM, Talos F200x, USA) were used to observe the surface morphologies of ATP, hole-cut ATP, and ULM@ATP. The dynamic light scattering particle size analyzer (zeta, zetasizer Nano ZS, UK) was used to characterize the zeta potential of ATP and ULM@ATP by a 4 mW He-Ne laser operating at a wavelength of 633 nm. The specific surface area and porosity analyzer (Autosorb iQ, USA) was used to analyze the pore size of ATP, hole-cut ATP, and ULM@ATP, which could be used to verify the successful preparation of the porous structure. The release behavior of ULM molecules from the ULM@ATP was evaluated using an Ultraviolet spectrometer (UV-vis, Lambda 950, USA) with a test range of 200–800 nm.

2.6. Stability test of ULM@ATP/Oil coating

The rheological behavior of ULM@ATP/Oil was evaluated by the rotating rheometer (HR-3, TA, USA). To explore the stability of ULM@ATP/Oil coating, the Canon digital camera (EOS 70D) was used to record the flushing test of the ULM@ATP/Oil on an aluminum plate inclined 45° vertically under different conditions.

2.7. Self-healing test of ULM@ATP/Oil coating

To explore the self-healing ability of ULM@ATP/Oil, we repeatedly scratched the composite coating in air and in 1 M NaCl (pH = 7), 1 M HCl (pH = 1), and 1 M NaOH (pH = 14) solution, respectively. The open circuit potential (OCP) was used to investigate the recovery time of ULM@ATP/Oil coating protection performance during each self-healing process. Meanwhile, the OCP was also used to test the self-healing time of the minor scratch artificially fabricated on the ULM@ATP/Oil coating.

2.8. Corrosion resistance test of ULM@ATP/Oil coating

The corrosion resistance test of different coatings was carried out using the Q235 carbon steel substrate ($1 \text{ cm} \times 1 \text{ cm} \times 1 \text{ cm}$) as a working electrode. Q235 low carbon steel, polished by 400 #, 800 #, and 1500 # sandpaper, then ultrasonically washed in ethanol, was used as the substrate for different coatings. The corrosion resistance behaviors of ULM@ATP/Oil coatings immersed in 3.5 wt.% NaCl solution was investigated using an electrochemical workstation (Gamry, USA). Electrochemical impedance spectroscopy (EIS) was recorded in the frequency range of 10^{-2} Hz to 10^5 Hz at 20 mV sinusoidal voltage signal amplitude. The thickness of the ULM@ATP/Oil was $1 \pm 0.2 \text{ mm}$. SEM and Raman were used to detect the corrosion morphology and chemical composition of corrosion products.

2.9. Antifouling test of ULM@ATP/Oil coating

The antibacterial ability of ULM and ULM@ATP was evaluated by the bacteriostatic ring test. A nutrient solution containing Escherichia coli was equably spread on agar plates. Drug-sensitive paper (6 mm diameter, circular disk) immersed in ULM solutions was positioned on agar plates and cultured in an incubator at 37°C for 24 h. Simultaneously, ATP and ULM@ATP powders were compressed into discs (14 mm diameter, circular disk), placed on an agar plate, and cultured in an incubator at 37°C for 24 h. Record the size of the bacteriostatic zone of the samples and evaluate the

bacteriostatic ability. The bare titanium sheet and the metal titanium sheet coated with ULM@ATP/Oil coating were placed in the chlorella solution at 37 °C for 48 h, then the antifouling ability of ULM@ATP/Oil was evaluated by bio-laser confocal (LEICA TCS SP5). The surface coverage of chlorella on different samples was measured by the Image Color Summarizer.

3. Results and discussion

3.1. Characterization of functionalized ATP

A series of structure characterizations of raw ATP, hole-cut ATP, and ULM@ATP were examined by SEM and TEM, as shown in Fig. 2. ATP was a naturally available one-dimensional nano-container with high surface area, moderate cation exchange capacity, and excellent salt resistance compared with other clay minerals [31]. In contrast to the layered clay minerals, the one-dimensional clay mineral of ATP had a unique rod-like crystal structure and contains more silanol groups and other active sites on the whole external and internal surface [32]. Surface properties of one-dimensional ATP were not just known for its larger specific surface area, it can be coupled with the characteristics of structural cavities and different types of water molecules [33], performing excellent nano-container for assembling advanced nano-structured functional materials [23].

ATP had a trioctahedral character with the theoretical structure formula of $(\text{Al}_2\text{Mg}_2)\text{Si}_8\text{O}_{20}(\text{OH})_4 \cdot 4\text{H}_2\text{O}$ for a half-unit cell [34]. As it was well-known, the natural ATP commonly aggregated together (Figs. 2(a) and S1(a) in the Supporting Information) due to the van der Waals forces and hydrogen bonding interactions among the nano-sized fibers [35], with fibrous micro-structure about 20–30 nm in diameter, 0.5–1 μm in length and about 3.7 $\text{\AA}^2$ –6.0 $\text{\AA}^2$ wide channels [36]. The raw structure affected the dispersion of natural ATP in the coating and greatly reduced the active adsorption sites, which could decrease the loading capacity toward ULM due to the small specific surface area. After hole cutting, the ATP crystal bundles were partially broken down and formed some piece-like aggregates (Fig. 2(b)), which indicated that the hole-cutting process could enhance the dissociation of ATP crystal bundles to a certain degree. After the hole cutting, the carbonate in the ATP structure channel was reduced and some nano-structures were dissolved at the same time [37], thereby increasing the pore volume of ATP [38]. The specific surface area and pore volume of ATP after ammonium fluoride cutting were effectively increased [39], which would be favorable for its loading capacity.

After loading of ULM, the ATP even broke most of the piece-like aggregates [35] and achieved obvious dissociation (Figs. 2(c) and S1(c)). This probably could be attributed to the repellent action of long-chain ULM adsorbed on the ATP surface and inside channels. Characteristics of the outer surface and interlamellar region had generally altered after hole cutting and modifying by ULM. The surface morphology and N element distribution in Fig. 2(d, e) illustrated that the ULM was loaded on the external and internal ATP uniformly (more element distribution can be found in Fig. S4). The mass content of the N element on the ULM@ATP obtained by the TEM-EDS test was 18.27 wt.% illustrating a huge amount of ULM was loaded on the ATP. The ULM was usually located at the edges of the channels (i.e. those tunnels acceding to the external surface of the silicate) which were directly accessible to reagents such as corrosion inhibitors [40]. The ULM modification would endow the ATP to reinforce the self-healing composite coating with active corrosion resistance capacity. The TG curves of the ATP before and after modification with ULM were analyzed under the protection of argon. As shown in Fig. 2(f), the obvious initial weight loss of pristine ATP in the range of 40–100 °C was attributed to the release of physically adsorbed water and inter-layer water [41]. The

subsequent weight loss from 80 to 700 °C was attributed to the removal of crystal water, zeolite water, and coordinated water in ATP, accompanied by the dihydroxylation reaction [42]. For ULM, the major weight loss appeared in the range of 160–280 °C, which was due to the disintegration of ULM. After modification with ULM, the weight loss of ULM@ATP was ascribed to the removal of different types of water and the decomposition of grafted ULM in ATP. Compared to the TG curves of ULM, hole-cut ATP, and ULM@ATP, the mass fraction of ULM grafted on the ATP was about 6.71 wt.%.

The FT-IR spectrum of raw ATP, hole-cut ATP, and ULM@ATP were shown in Fig. 2(g). In the FT-IR spectra of the ATP powders, the absorption bands at 3263, 3394, and 1659 cm^{-1} were ascribed to the –OH stretching and bending vibrations of ATP [41], and the absorption bands at 1446 cm^{-1} were attributed to the Si–OH groups of the ATP [43]. After modification with ULM, except for the –OH and Si–OH groups, new peaks appeared at 2925 and 2857 cm^{-1} corresponding to C–H stretching vibration on methylene and the methyl of alkyl groups [44] and 1402 cm^{-1} (the bending of CH_2) in the spectra of the modified ATP [41], indicating that the ATP was successfully modified with the ULM. As shown in Fig. 2(h), the XRD patterns of the ATP showed little change after modification. It demonstrated a characteristic diffraction peak (110) at 8.5° of the ATP, revealing that the structure of ATP was maintained after the hole cutting and modification treatment. The weak reflections at 13.8° (200), 19.9° (040), and 27.5° (231) could be observed, indicating the structural integrity of ATP [45].

As known, the zeta potential was usually used to illustrate the adsorption and dispersity of fillers in the coating. In this work, after hole cutting, the zeta potential was changed because the octahedral metal ions in ATP were gradually removed. The removal of metal cations in octahedral sheets by the ammonium fluoride cutting made more Si–O–M bonds broken as Si–O bonds, leading to the rise of negative surface charges on the fibrous ATP [46], and enhancing the capacity for loading functional groups. After being modified by ULM, the zeta potential was negatively charged from 9.26 mV (Fig. S2) to –24 mV (Fig. 2(i)). ULM modification altered the surface charges [47] and promoted the formation of hydroxyl groups on the ATP [48], which could increase the dispersion uniformity of the ULM@ATP nanoparticle in the coating.

The nitrogen adsorption-desorption curve in Fig. 2(j–l) was used to measure the surface area, pore volume, and porosity [49] of ATP before and after modification with ULM. The isotherm was of H3 type according to the IUPAC classification. Meantime, the hysteresis loop for the raw ATP was narrow and small, suggesting only a small part of mesopores existed in the nano-container with a surface area and pore volume of 102.74 $\text{m}^2 \text{g}^{-1}$ and 0.397 $\text{cm}^3 \text{g}^{-1}$, respectively. After hole cutting by ammonium fluoride (Fig. 2(k)), the hysteresis loop become wider and higher, illustrating the porosity of the ATP was increased, and the surface area and pore volume increased to 126.255 $\text{m}^2 \text{g}^{-1}$ and 0.615 $\text{cm}^3 \text{g}^{-1}$, respectively. ATP was known for its porous structure and could act as a nano-container, the hole cutting further enhanced the porosity of ATP, which would significantly increase its adsorption ability. However, the surface area and pore volume of ATP decreased after modification by ULM (106.679 $\text{m}^2 \text{g}^{-1}$ and 0.401 $\text{cm}^3 \text{g}^{-1}$) for the reason that the active sites on ATP after hole cutting were reacted with ULM. The nitrogen adsorption-desorption curve further indicated the hole cutting and modification process of ATP which results were well consistent with the FT-IR, XRD, zeta potential, and TEM-EDS results, these results drew a common conclusion that the ULM was successfully loaded on the ATP.

It was well-known that the adsorption capacity of ATP depends on its porous microstructure. Hole cutting of ATP could not only continuously leach the metal cations in the octahedral sheets but also decompose the SiO_4 tetrahedral framework. Moreover, the leaching amount of Si^{4+} was much larger than that of Al^{3+} , Mg^{2+} ,

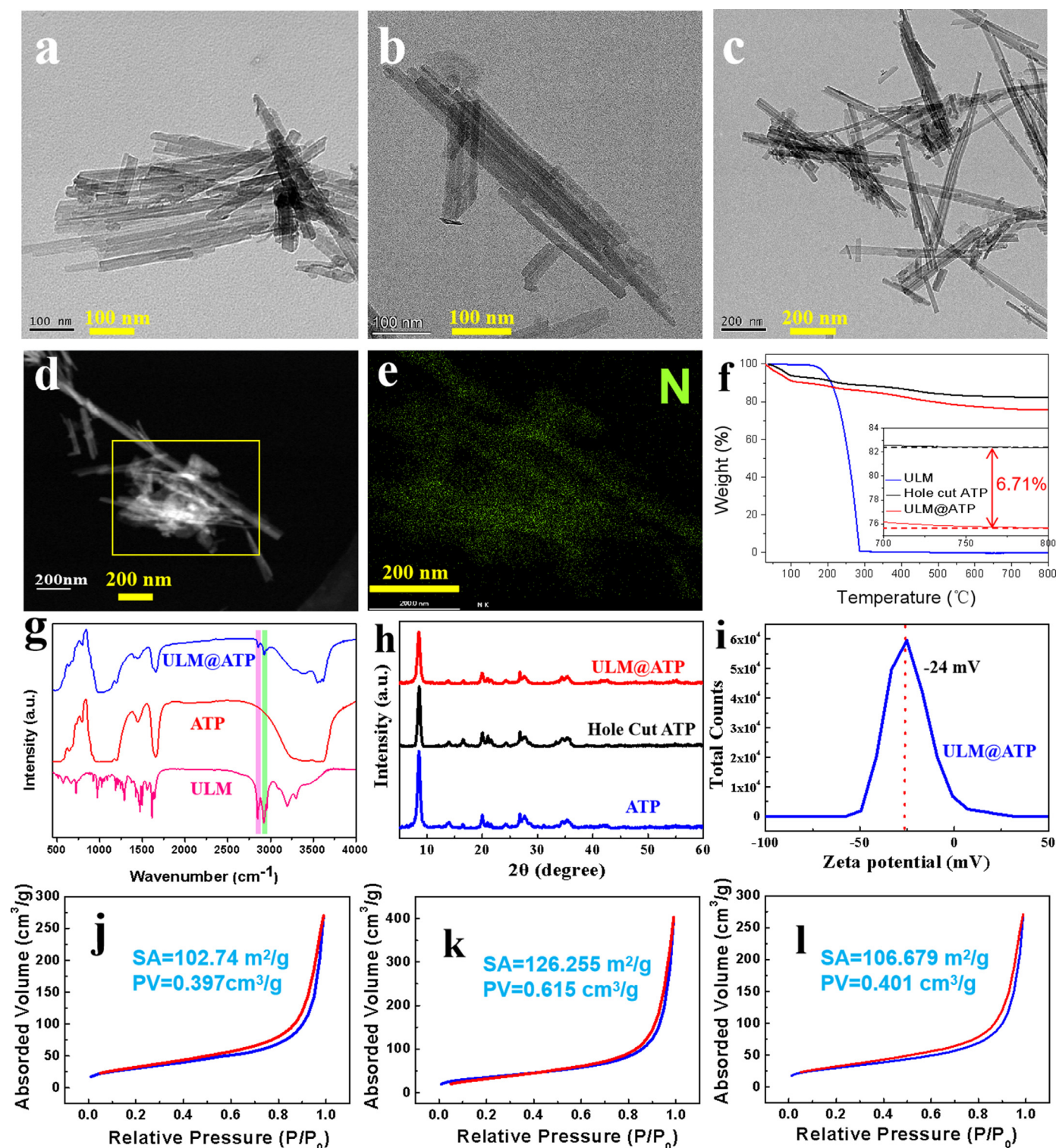


Fig. 2. TEM images of (a) raw ATP, (b) hole-cut ATP, (c) ULM@ATP, (d) dark field image of ULM@ATP, (e) N element distribution of (d), (f) TG curves of ULM, hole-cut ATP, and ULM@ATP, (g, h) FT-IR spectrum and XRD of raw ATP, hole-cut ATP, ULM@ATP, and (i) zeta potential of ULM@ATP, and nitrogen adsorption-desorption curves of raw ATP (j), (k) hole-cut ATP, and (l) ULM@ATP.

and Al ions could be more easily leached than Mg ions, meaning that the dissolution of octahedral cations was no longer dependent on the occupation mechanism but may be related to the strength of the binding energy [50]. In this study, the synthesis of ULM@ATP was a key part of our strategy for the fabrication of dual-function self-healing coating. The unique structure, morphology, and texture characteristics provided ATP with excellent carrier

performance, adsorption performance, and reinforcement performance [51]. Hole cutting improved the adsorption capacity of ATP to ULM inhibitor, it was indispensable to disaggregate ATP crystal bundles for improving the porosity of ATP and increasing its loading ability. The ULM loaded on the ATP enhanced the dispersion of ATP in the coating and could form an adsorbed film on the metal surface to resist the ingressing of the corrosive medium.

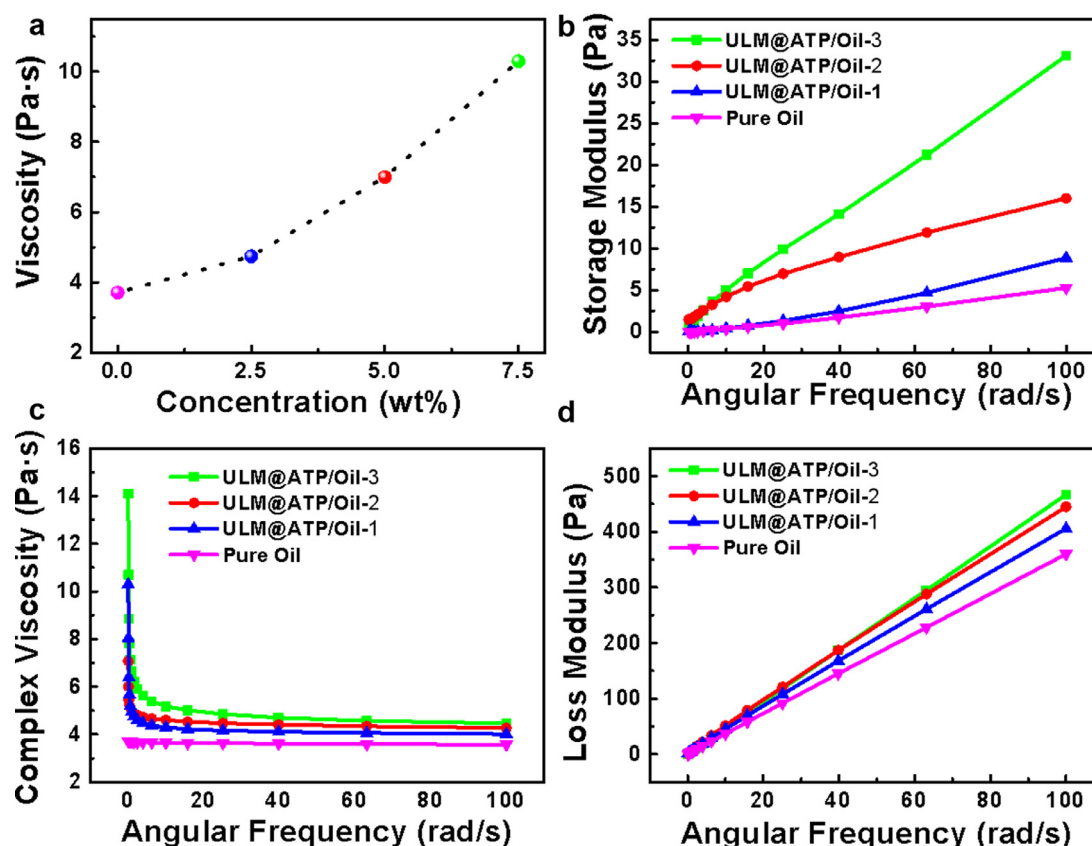


Fig. 3. The viscosity (a), storage modulus (b), complex viscosity (c), and loss modulus (d) of ULM@ATP reinforced oil composite.

3.2. Viscosity and rheological properties

As can be known in our previous work [5,52], the viscosity and rheological properties of the fluidity-driven self-healing materials was a key role in their high self-healing efficiency. With the increase of carbon nanotube [5] and graphene oxide [52], the viscosity of the composite enhanced, the self-healing time was prolonged and the corrosion resistance increased. Therefore, the self-healing efficiency and the functions (such as anti-corrosion and anti-biofouling capacity) were two contraries of coating. The ATP modified by ULM endowed the anti-biofouling and active corrosion protection properties to the self-healing coating. On the one hand, the fluidity of ATP-reinforced oil coating was decreased with the increase in ATP content. The mobility of the oil matrix promoted the healing of the damaged area, as well as the reformation of the broken bonds [4]. If the mobility of the composites was not adequate, it would not flow to recover the damage or it would do so slowly. On the other hand, the increase of ULM@ATP content could enhance the anti-corrosion and anti-biofouling properties of the composite coating which showed a wide potential application. The challenge was to balance the self-healing efficiency and dual-function properties with an appropriate content of thickening fillers (ULM@ATP).

Stability is the key role of fluidity-driven self-healing material in the corrosive solution for some immersion service environments. As shown in Fig. 3(a), the stability, as well as the viscosity (Fig. 3(a, b)), was increased with the content of ULM@ATP because the functionalized ATP with the modification of ULM, formed physically intertwined structures in the oil matrix [53]. With the concentration of ULM@ATP increased or the angular frequency decreased, the complex viscosity of the composite fluid coating increased (Fig. 3(c)), and the stability of the composite coating was

enhanced even by the rinsing of high-speed water, (Fig. S5), 1 M HCl solution (Fig. S6), 1 M NaCl solution (Fig. S7), and 1 M NaOH solution (Fig. S7), respectively. This result illustrated the shear-thinning behavior of the ULM@ATP/Oil fluid composite. The shear-thinning behavior may be due to the entanglement of fibrous ATP in the oil, which required longer relaxation time and higher shear stress for the high-viscosity composite oil to flow [54]. The elastic behavior of ULM@ATP/Oil depended on the energy storage mechanism at the filler/oil interface. The relaxation of the ULM@ATP network was longer than that of the individual oil chains [55] and it was tardy when compared with the pure oil which was due to the incorporation of the physically intertwined network of ULM@ATP. The interaction of these networks limited the long-range motion of the oil chains and resulted in slight gel-like behavior, thus, the curves in Fig. 3(d) for the ULM@ATP/Oil composites became steeper [56]. The viscosity and rheological properties of composite coating provided a guide for adjusting the appropriate filler content to balance the self-healing efficiency and dual function of the composite coating.

3.3. Self-healing efficiency and anti-corrosion performance

The self-healing procedure in the air (Fig. 4(a)), in 1 M HCl solution (pH = 1, Fig. 4(b)), 1 M NaCl solution (pH = 7, Fig. 4(c)), and 1 M NaOH solution (pH = 14, Fig. 4(d)), showed a huge improvement compared with conventional self-healing coating which mostly healed the damage only in the air [5]. Two aspects contributed to the composite coating at the damage and self-healing processing in solution service surroundings. On the one hand, the fluidity of the composite coating could heal the damage in several or tens of seconds. The self-healing led the composite coating more meaningful when used under immersion conditions, especially in

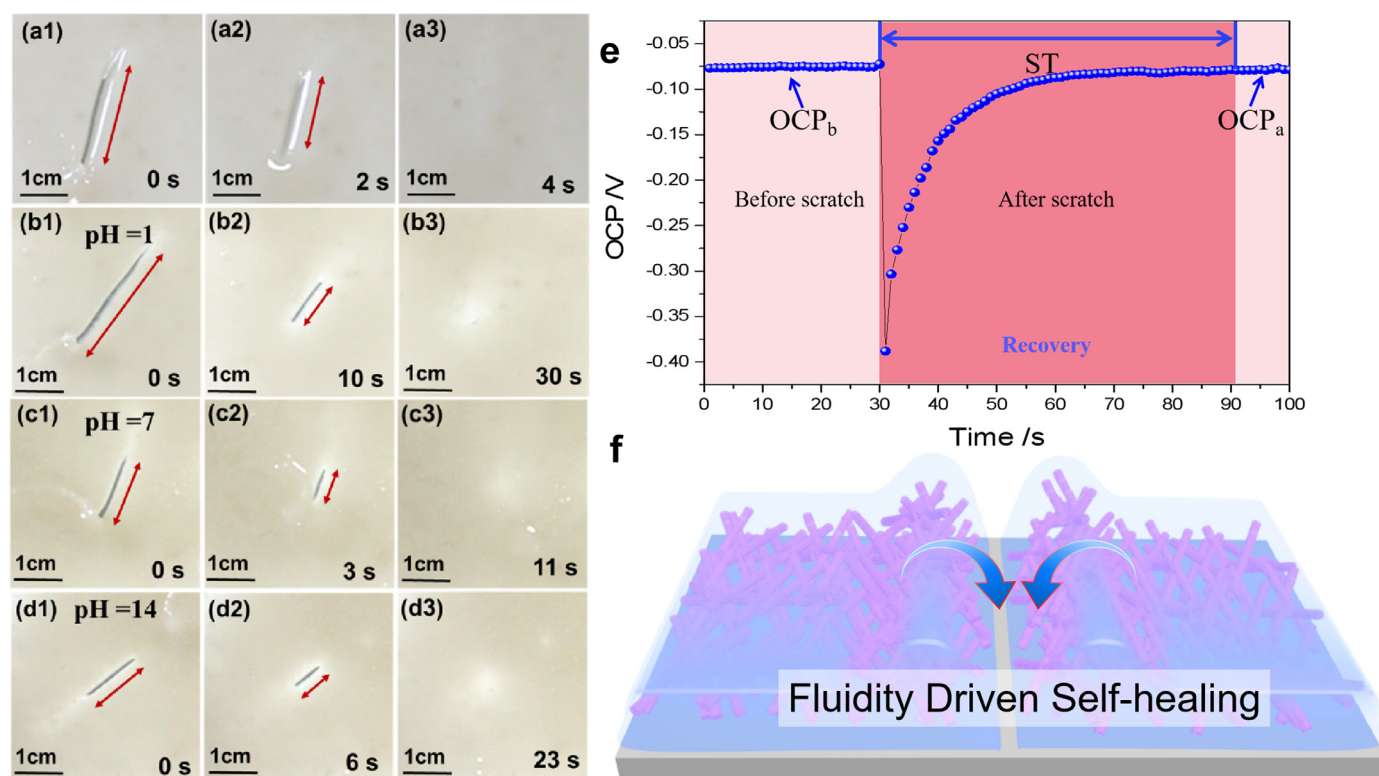


Fig. 4. The self-healing process of ULM@ATP/Oil coating in the air (a), in a solution of (b) 1 M HCl, pH = 1, (c) 1 M NaCl, pH = 7, (d) 1 M NaOH, pH = 14, (e) OCP of ULM@ATP/Oil coating with continuous damage at the same position, (f) schematic representation of the self-healing mechanism of ULM@ATP/Oil coating.

acid and alkaline environments. On the other hand, the enhanced viscosity of the composite improved the adhesion of the coating on the metal substrate, and residual coating on the metal interface separated the metal from corrosive solutions when the damage occurred [5]. In the air, the self-healing was easy, however, the self-healing was impeded in solution, and the healing efficiency was decreased. According to the harsh aggressiveness of the HCl solution, the hydrogen bubble may be formed by the reaction between HCl and metal substrate which would impede the flow of the composite to the damaged region. Thus, the self-healing efficiency of the composite coating in 1 M HCl solution was the lowest which spent about 30 s. But by and large, the self-healing efficiency of the composite coating was much higher than traditional solid self-healing materials. The self-healing characterization in solution especially in acid and alkaline environments was rarely and meaningful for improving the service range of the self-healing materials.

To further study the self-healing efficiency and active corrosion protection performance of the composite coatings, the OCP of the composite coating with continuous damage was investigated (Fig. 4(e)). As can be known from our previous work, the self-healing ability of the fluidity-driven self-healing material was infinite due to the endless mobility of the fluid matrix [5]. The OCP was usually used to evaluate the corrosion resistance of coatings, while in this work, the OCP was used to investigate the self-healing efficiency of ULM@ATP/Oil composite coating. The OCP was measured during the continuous damage and self-healing process. A sharp decrease in the potential appeared immediately after scratching. Then the OCP gradually upturned to the value for OCP_a (OCP after self-healing) after healing. The ST (self-healing time) for different coatings was shown in Table 1. As the ULM-modified ATP was added to the coating, the self-healing time increased with the viscosity of the composite coating (Fig. 3). The scratch and self-healing of ULM@ATP/Oil-2 coating were repeated 56 times in total, taking 3420 s, with an average of ~61 s. All 56 times self-

Table 1

The OCP before and after scratch, and the average self-healing time of the composite film after scratch.

Samples	OCP-b (V)	OCP-a (V)	Average ST (s)
Oil	-0.195	-0.223	55
ATP/Oil-1	-0.152	-0.165	56
ATP/Oil-2	-0.086	-0.090	60
ATP/Oil-3	-0.080	-0.085	80
ULM@ATP/Oil-1	-0.128	-0.084	58
ULM@ATP/Oil-2	-0.075	-0.080	61
ULM@ATP/Oil-3	-0.077	-0.081	82

healing still possessed a short ST in the level of tens of seconds to 100 s. The complete self-healing time of ULM@ATP/Oil coatings (Table 1) was lower than the self-healing materials driven by hydrogen bond, Light, π - π Stacking, etc. Moreover, the corrosion resistance of ULM@ATP/Oil coatings was higher than previously reported self-healing coatings [5,19,52,57,58]. The complete recovery of the OCP for the composite coating illustrated the scratch on the coating was self-healed and the corrosion resistance of the coating was fully recovered. The ULM@ATP could form a adsorb film on the metal surface when the damage occurred and the active corrosion protection during the self-healing process was positive to the recovery of the coating's corrosion resistance. When repeated the scratches on the composite coating, the OCP was fully recovered continuously 56 times at the same position. For traditional self-healing materials, the self-healing was just worked single time with a low self-healing efficiency (long self-healing time and low function recovery) at the same position. The continuous damage and self-healing indicated the composite coating possessed a potential of infinite self-healing ability, which showed a wide application in the future multi-function coating in more serviced surroundings.

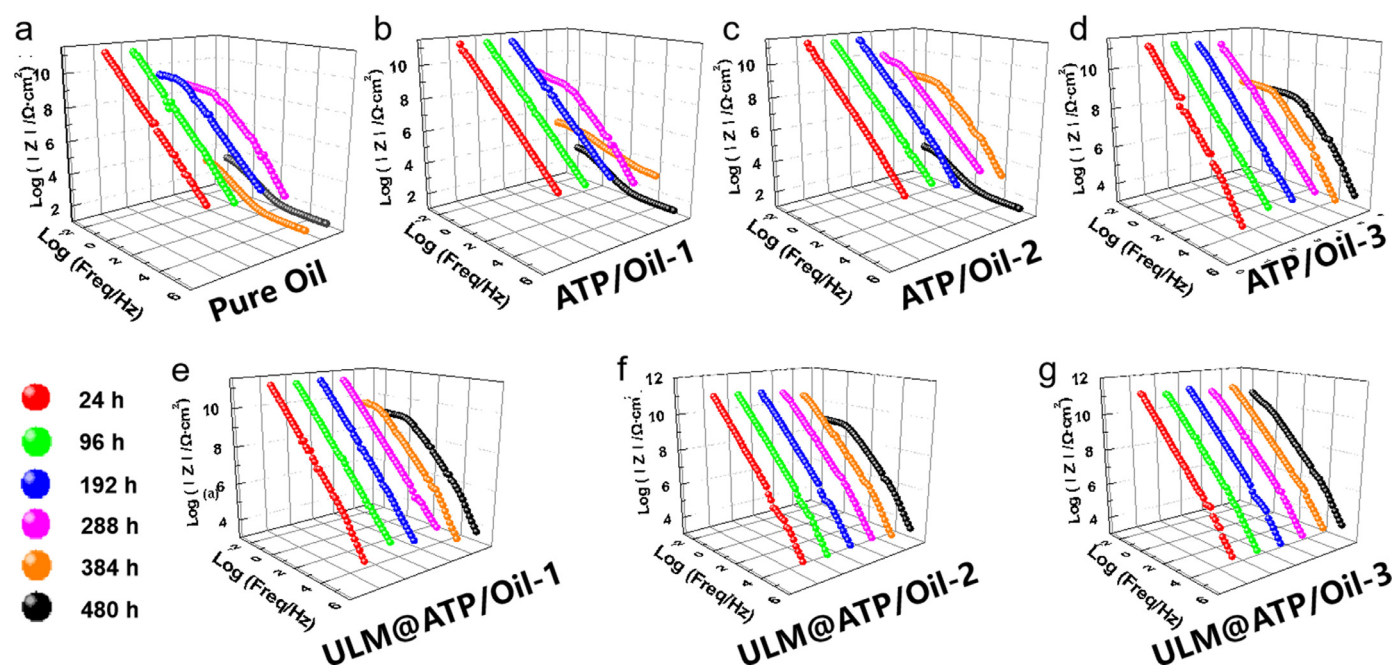


Fig. 5. The impedance modulus of pure Oil coating (a), ATP/Oil coatings (b–d), and ULM@ATP/Oil coatings (e–g) immersed in 3.5 wt.% NaCl solution for 480 h.

Fluid material can recover damage rapidly without external triggering (Fig. 4(f)) in contrast with the traditional self-healing material which presents low healing efficiency and external triggering demand. The dual-functional self-healing coating in this work was fabricated by thickening of ULM@ATP, the self-healing efficiency directly consisted of the composite viscosity. The ULM-modified ATP played a complex role in the self-healing efficiency and the coating function was changed in the contrary direction. Moreover, with the increase of ULM@ATP, the stability of the self-healing material increased which was important for the corrosion protection coatings used in harsh immersion environments. Secondly, the anti-corrosion and anti-biofouling properties of the self-healing coating would be enhanced due to the addition of ULM@ATP. To balance the stability and the self-healing efficiency, the content of ULM@ATP was important to be controlled.

In the ULM@ATP/Oil composite coating, the dispersion of ULM@ATP also lengthened the diffusion path of the corrosive species and delayed the water intake. Furthermore, the ULM loaded on the ATP would highly enhance the inhibition of corrosion because the ULM could adhere to the metal surface when a scratch occurred on the coating. The ULM was known for its corrosion inhibition and the ATP acted as a nano-container that caught the amount of ULM (showed by FT-IR, XRD, and DES characterization), which performed excellent corrosion protection behavior during the self-healing process.

The long-term corrosion resistance of the ULM@ATP/Oil coatings was investigated by 480 h immersion in 3.5 wt.% NaCl solution. As shown in Fig. 5, the impedance modulus of pure Oil, ATP/Oil, and ULM@ATP/Oil coatings was reduced with the prolonged immersion time. As the content of ATP increased, the viscosity, as well as the stability of the composite coating enhanced, meanwhile, the corrosion resistance of the composite coating increased. After the modification of ATP with ULM, the corrosion inhibitor ULM could form an adsorbed film leading to the active corrosion protection of composite coating before the chloride ions permeated into the interface at the metal substrate. Furthermore, the resulting homogenized dispersion of ULM@ATP in the oil matrix could remain stable for a long time without gravity sedimentation, which could be attributed to the skeleton effect of ATP and

the hydrogen bonding effect among the ULM@ATP and oil matrix. The ULM@ATP/Oil-2 coating performed the highest impedance modulus of $1.65 \times 10^{11} \Omega \text{ cm}^2$ and was higher than previously reported self-healing coatings (Table S1) which was driven by the hydrogen bond and the release of Corrosion inhibitors [57,58].

After 480 h immersion in the 3.5 wt.% NaCl solution, the impedance modulus of ULM@ATP/Oil was still maintained at a high value which illustrated the ULM@ATP significantly improved the corrosion resistance of the composite coating. In contrast, the ATP/Oil coating was gradually degraded under the attack of corrosive species, which only serves as a adsorb barrier without an active protection function [59]. The impedance modulus of ULM@ATP/Oil was about 1–3 magnitude of orders higher than that of pure Oil coating and ATP/Oil coatings after 480 h immersion in 3.5 wt.% NaCl solution, demonstrating an active corrosion protection effect. The surface of natural fibrous clay was constituted by a great number of silanol groups [31], which were reactive and can adsorb massive inhibitors (ULM). According to the multi-experimental verification, ATP clay constituted an interesting material to be used as a nano-container of inhibitor hindering the corrosive species to permeate into the coatings. Moreover, the protection mechanism was also due to the trapping of chloride anions loaded on ATP via the exchange with ULM, which led to the inhibition of pitting corrosion. The availability of aggressive species such as chloride ions or other species during the corrosion process provided triggering conditions for the release of ULM. The release mechanism was based on an exchange process between adsorbed inhibitors (ULM) and the exchangeable aggressive species (such as chloride ions) [60].

Moreover, two factors were beneficial to protecting the metal substrate by the ULM [61]. On one hand, the exposed Fe would be covered by the imidazoline ring and heteroatoms, consequently preventing the metal from corrosion. On the other hand, as a unipolar tail, the alkyl chain prevented water molecules from contacting the metal substrate. The 3D crosslinking network of ATP enhanced the stability of the composite coating under the immersion environment and the adsorbed film formed by ULM performed excellent active corrosion resistance for the composite coating in long-term corrosion protection.

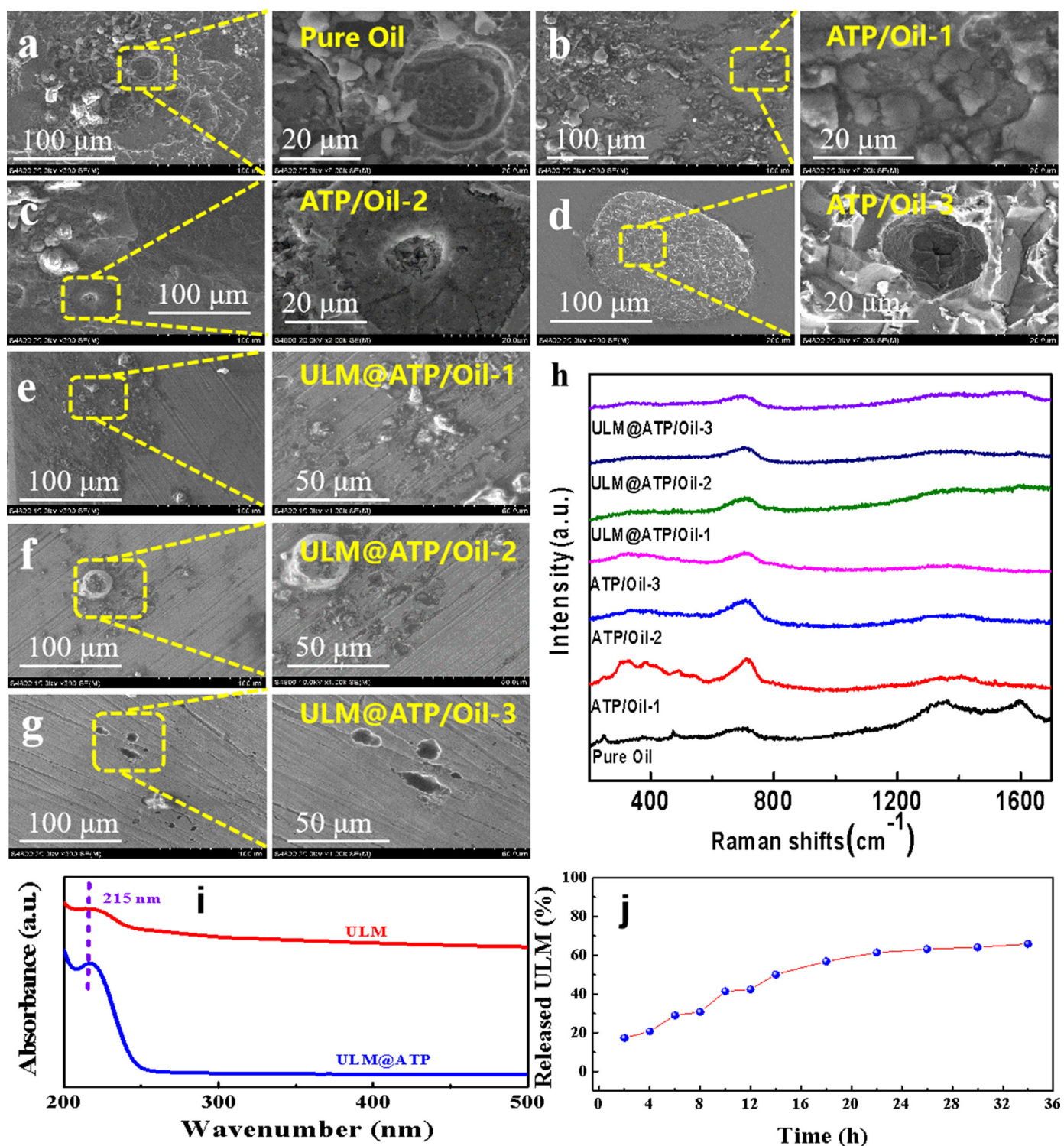


Fig. 6. SEM images of the metal substrate under the protection of (a) pure Oil, (b–d) ATP/Oil coatings, (e–g) ULM@ATP/Oil coatings, (h) Raman spectrum of the corrosion products on the metal substrate protected by different coatings, (i) the UV–Vis of ULM and ULM@ATP, and (j) release curves of ULM@ATP in the 3.5 wt.% NaCl solution.

After 480 h immersion in the 3.5 wt.% NaCl solution, the corrosion degree of metal substrates under the protection of different coatings was shown in Fig. 6 by the characterizing of corrosion products on the metal surface. Many corrosion products and serious pitting corrosion appeared on the metal substrate under the pure Oil coating (Fig. 6(a)), which indicated the corrosion resistance of pure Oil was inadequate for the long-term protection of metal. For the ATP-reinforced composite coatings (Fig. 6(b–d)), the

corrosion products and the pitting corrosion on the metal substrate were reduced compared with the pure Oil coating, which was due to the barrier properties of the crosslinking network of ATP in the oil matrix. For ULM@ATP/Oil coatings (Fig. 6(e–g)), there were just a few corrosion products on the metal surface. Even the polish grooves left by sample preparation could be observed clearly, which illustrated the corrosion on the metal substrate was alleviated under the protection of the ULM@ATP/Oil coatings.

The corrosion products on the metal substrate measured by Raman further investigated the corrosion level and the protection ability of different coatings. The metal under the pure Oil coating was performed in the later corrosion stage, with four kinds of corrosion products appearing on the surface. The band at 245, 471, 692, and 1348 cm^{-1} represented $\gamma\text{-Fe}_2\text{O}_3$, $\alpha\text{-FeOOH}$, Fe_3O_4 , and $\beta\text{-FeOOH}$, respectively [62]. According to the barrier of the ATP crosslinking network, the types of corrosion products for ATP-reinforced composite coating were reduced. The band at 318, 485, and 715 cm^{-1} correspond to the $\alpha\text{-FeOOH}$ and Fe_3O_4 . There was just one kind of corrosion product (Fe_3O_4) that can be found on the metal substrate under the protection of ULM@ATP/Oil coating. The SEM and Raman characterization of the corrosion products on the metal substrate under the protection of different coatings optically and directly presented the corrosion resistance of the coatings. According to the enhanced stability of the composite coating and the active corrosion resistance of ULM, the ULM@ATP/Oil coating performed the best anti-corrosion performance.

To avoid undesirable leaching and enhance the inhibition efficiency of the inhibitor, the controllable release was necessary for an active anti-corrosion system. The loading ability and release property of hole-cut ATP were performed by UV-Vis and release curves. In Fig. 6(i), the characteristic absorption band at 215 nm illustrated the ULM was loaded on ATP. The release curve of ULM@ATP in the 3.5 wt.% NaCl solution was used to investigate the release behavior of ULM in the corrosive environment. For the early 18 h, the release of ULM mostly came from the outer adsorption of ULM on ATP, and the release speed was high. After the release for 22 h, the release speed decreased and the released ULM originated from the internal surfaces or some physical filling in the ATP. ULM inhibiting the corrosion of steel substrate was an active process. With the penetration of corrosive species, ULM was released to the surface of the steel substrate combined with iron ions and formed a dense passivation film on the metal substrate surface [63] to inhibit the further corrosion of the metal substrate. Moreover, the long-term immersion test and the release curve confirmed that the release of corrosion inhibitor endowed a higher protective ability and increased barrier properties of the ULM@ATP/Oil coating.

The release behavior of ULM@ATP in Fig. 6(j) showed that the loading of ULM could be continuously and stably released under specific conditions and stimulation of corrosive species which maintained excellent anti-corrosion performance for a long time [64]. Therefore, the filler of ULM@ATP added in fluid self-healing coating was more efficient than added in the common solid anti-corrosion coating. The release of ULM in the fluidity self-healing coating was easier and the adsorbed film on the metal substrate would renovate as the damage occurred, which can be verified by the continuous damage and self-healing process in the OCP test. The active protection was meaningful within the self-healing process, and the recovery of the corrosion resistance of the composite coating illustrated the quick stimulus-response and controllable release performance of ULM when the damage occurred.

3.4. Anti-biofouling properties

The antibacterial ability of ULM and ULM@ATP was evaluated by the bacteriostatic ring test. A nutrient solution containing *E. coli* was equably spread on agar plates. ATP was not only effective in fabricating anti-corrosion coating but also can be easily engineered to form an anti-biofouling composite through the introduction or modification of organic antibiotic constituents [23] (ULM). In Fig. 7(a, b), dipped the drug-sensitive paper with a diameter of 6 mm was dipped into ULM solution, then these were placed on the agar plate containing *E. coli* bacterial nutrient solution, cultured in a 37 °C incubator for 24 h, and recorded the radius of the an-

tibacterial circle to evaluate the antibacterial effect. The radius of the antibacterial circle was about 16.5 mm for 24 h, which meant ULM was an ideal anti-bacterial material. To further investigated the anti-bacterial performance of ULM@ATP (Fig. 7(c, d)), the ATP and ULM@ATP powder were pressed into the sheet (with a diameter of 14 mm), and these sheets were immersed into the *E. coli* bacterial nutrient solution for 24 h. An obvious antibacterial circle with a diameter of 55.5 mm appeared which was attributed to the anti-bacterial of ULM@ATP. For comparison, a slight antibacterial circle appeared (23.3 mm) for the ATP sample, which illustrated the antibacterial performance of the ULM@ATP sample mostly originated from the ULM.

In terms of the antimicrobial actions, one hand can be considered that once the ULM was adsorbed on the cell membrane of the bacteria, it would affect the cell permeability and prevent normal material exchange between the environment and the bacteria, showing a bactericidal effect [65]. Another bacteriostatic action may be caused by the needle stab effect from the fibrous ATP damaging the cytoplasmic membrane of the cell [66]. Furthermore, the hole-cut ATP exhibited improved surface activity which enhanced the interfacial interactions between the fibrous ATP with ULM. Besides, the pretreatment of ATP was of great significance to prepare safe and effective antibacterial ULM@ATP nanofillers, the green and simple modification showed a wide application potential.

The algae resistance of pure Oil and ULM@ATP/Oil coatings were investigated, and the optical images and bio-type laser images of the chlorella settled on the coatings were shown in Fig. 7. In general, the algae were almost completely covered on the pure titanium surface, and fewer algae were attached to the surface of pure Oil coating which indicated silicon oil could resist the attachment of algae to the oil surface. The silicone oil replaced the solid surface (Ti) to contact the algae, so algae were difficult to attach to the surface under the shearing force of water due to the super-slip characteristics of the oil surface [67]. At the same immersion condition, fewer algae adhered to the ULM@ATP/Oil coating surface. Meanwhile, the algae attachment was reduced significantly with the increase of the ULM@ATP content. Green chlorella solution in Fig. 7(e, f) indicated the chlorella survived healthy in the test condition, however, the chlorella was hard to see in the ULM@ATP/Oil sample plate, especially in Fig. 7(h, i). This phenomenon illustrated that the chlorella was killed by the released ULM from the coating into the nutrient solution. More intuitively, a number statistics graph was made to show the number of adhered algae. The surface coverage in Fig. 7(j) showed the chlorella adhered to pure Oil and ULM@ATP/Oil coatings decreased by 54.76%, 55.36%, 78.23%, and 96.88%, respectively, compared with the pure Ti surface. When compared with pure Oil, the chlorella adhesion of ULM@ATP/Oil was reduced by 1.32%, 51.89%, and 93.1%, respectively.

On one hand, the strong adsorption property as well as the rod-like or fiber-like crystals that may cause the “needle stab effect”, can adsorb bacterially and damage the cytoplasmic membrane of the cell, producing antimicrobial effects [66]. On the other hand, ATP had the capacity as a carrier for antimicrobial agents. Generally, the small molecules can freely access the tunnels and channels [21], and the other compounds may be loaded on the external surface through interactions with the silanol groups on its outer surface [68,69]. The particular nanosized and active surface groups of clay minerals make them potential to be applied as nanofiller to improve the comprehensive performance of polymer materials, such as anti-corrosion [35] and anti-bacterial [66].

3.5. Self-healing strategy and active corrosion protection mechanism

The ATP was widely applied in the construction of improved nanocomposites or hybrid materials properties due to its large specific surface area, surface-active groups, porosity, and surface

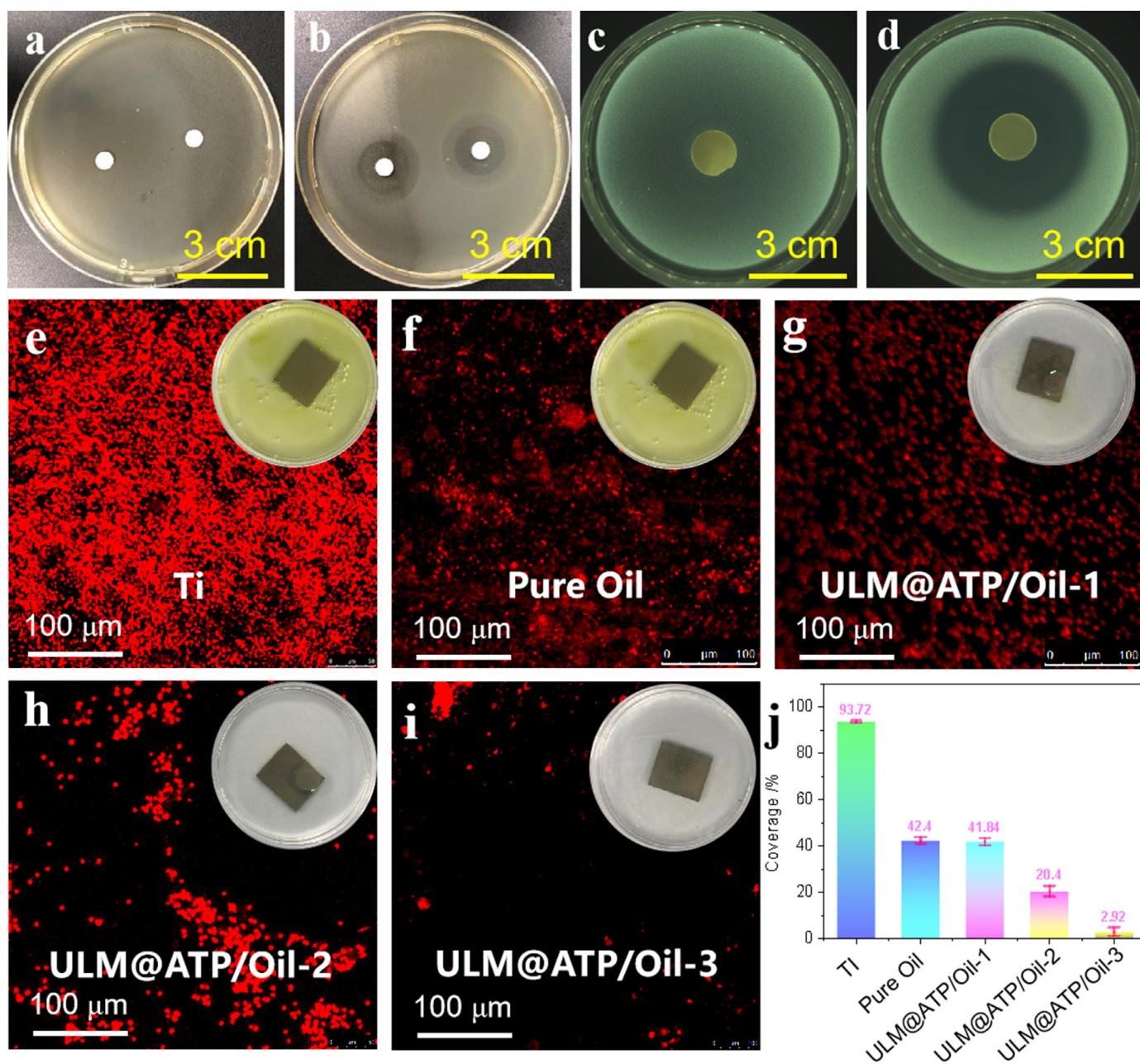


Fig. 7. The antibacterial ability of ULM in *E. coli* bacterial nutrient solution for (a) 0 h, (b) 24 h, the antibacterial ability of ATP (c), and ULM@ATP (d) powder sheet (with a diameter of 14 mm) in *E. coli* bacterial nutrient solution for 24 h, the optical images and bio-type laser images of the chloroform settled on the Ti matrix (e), (f) pure Oil, (g) ULM@ATP/Oil-1, (h) ULM@ATP/Oil-2, and (i) ULM@ATP/Oil-3, and (j) the surface coverage of adhered algae on different coatings.

charge, which were derived from the unique structures and morphologies. The functional groups that existed on the external surface of ATP played an essential role in the interaction with the active protector ULM (Fig. 8). In addition, the strong adsorption property combined with the ULM played the antimicrobial properties, which was favorable to the enhanced antimicrobial activity of the nanomaterials. For the ULM@ATP/Oil coating, dual-action was essentially developed with a potential for several functionalities [58]: (1) released corrosion inhibitors (ULM) from ATP which could passivate the metal surface; (2) ATP network in the coating enhanced its stability which was important for the fluid self-healing material using under the immersion environment; (3) ULM loaded on ATP endowed the composite coating with active corrosion protection and ideal anti-biofouling properties. The natural clay mineral ATP was a promising material for thickening fluid self-healing ma-

terials, as the ATP presented excellent loading abilities, and was eco-friendly, cost-effective, and available in large quantities.

Application of the dual-action self-healing coating on metal surfaces had witnessed intensive previous research as one of the new generation technologies seeking to curb the menace of corrosion. However, there were certain challenges or limitations associated with self-healing coatings. For example, developing active anti-corrosion of self-healing coating with high efficiency (rapid healing and full recovery of properties) and other functions. Herein, significant potential for better corrosion protection is being proposed in multi-action self-healing coatings, displaying a combination of different healing mechanisms and dual-functionality. Because of the benefits associated with active corrosion protection and anti-biofouling capabilities of self-healing coating, combining the two functions into the same coating would bring about sev-

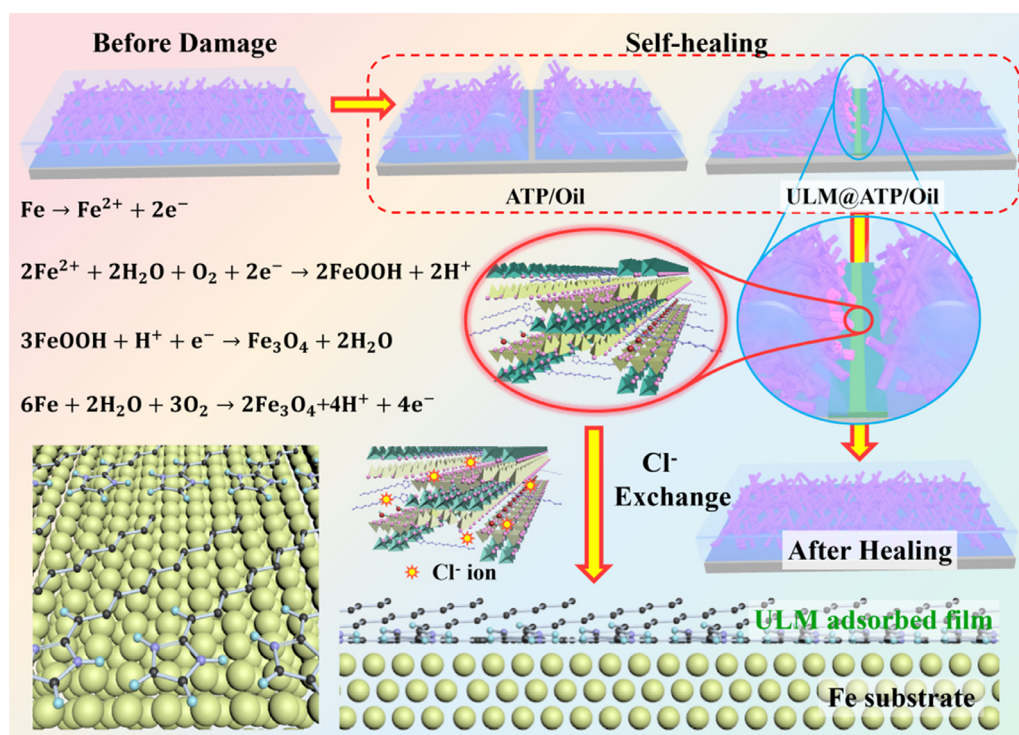


Fig. 8. Schematic representation of the self-healing strategy and active corrosion protection mechanism of ULM@ATP reinforced oil coating.

eral useful features for fluidity-driven self-healing material, including (1) rapid self-healing with high efficiency; (2) controlled releases of inhibitor as induced by natural nano-container; (3) active corrosion protection in the self-healing processing; (4) ideal anti-biofouling functions using in the marine environment.

Furthermore, once the coating is mechanically damaged, the metal substrate would not be exposed to corrosive species due to the adhesion of the fluid composite coating on the metal substrate. (5) Even the bulge of the rough metal surface exposed to the corrosive solution, the adsorbed film formed by the active function of ULM would further protect the metal substrate (Fig. 8). ULM can adsorb on the Fe surface through the imidazoline ring, by donating electrons to Fe atoms and accepting electrons from 3d orbitals of Fe atoms [59]. The composite material flowing back, the ATP network restructuring, and the active protection of inhibitor endowed a high corrosion resistance to the composite coating, even after continuous damages, exhibiting huge advantages for long-term application in harsh environments.

4. Conclusions

In summary, a dual-functional rapid responding self-healing coating was proposed for the synthesis of smart coatings to improve the active anti-corrosion and anti-biofouling integrated properties. The OCP test indicated the ULM@ATP reinforced oil coating displayed an infinite self-healing potential. The multi-action self-healing coating presented a superior active corrosion protection effect (impedance modulus was one to three magnitudes high than that of un-modified coatings) and anti-biofouling properties (anti-bacterial and algae resistance). These excellent properties and performance were raised from the following facts: (1) the ATP nanorods with abundant interconnected micro-/nanosized channels provided a large space to accommodate more ULM after hole cutting; (2) the endless fluidity of the oil matrix can fast heal the damage with continuous damage and provided active corrosion protection during the self-healing process; (3) the loaded ULM

performed excellent active corrosion protection and ideal anti-biofouling performance. The rapid responding self-healing coating with active anti-corrosion and anti-biofouling properties showed a wide application field in harsh marine environments.

CRediT authorship contribution statement

Yinghao Wu: Conceptualization, Methodology, Writing – review & editing. **Jinming Wei:** Data curation, Investigation, Writing – original draft. **Wenjie Zhao:** Supervision, Validation.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jmst.2022.11.029](https://doi.org/10.1016/j.jmst.2022.11.029).

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